

What is Paleoethnobotany?

Paleoethnobotany is the study of plant remains from archaeological contexts. Although most parts of a plant can be preserved archaeologically under specific conditions, most plant materials decay over time under normal conditions, leaving little archaeological evidence. The **taphonomic** conditions under which a particular plant, or plant part, entered the archaeological record and was preserved can also affect its likelihood of survival into the present. In general, plant remains are best preserved under completely waterlogged conditions, when frozen (e.g., in a glacier), or in extremely dry environments (e.g., the Atacama Desert of Chile or the Taklamakan Desert of Xinjiang, western China). Under more regular conditions, different categories of botanical remains are preserved under different conditions.

Paleoethnobotany is the study of plant remains from archaeological contexts. A synonym for this field of study is archaeobotany, a term generally preferred in Europe, while paleoethnobotany is more common in North America.

Archaeologists divide botanical remains into two categories: **macroremains** and **microremains**. Macroremains can be seen with the naked eye and include various seeds, fruits, and pieces of wood. Microremains are smaller and cannot be seen without the aid of a microscope. They include pollen, starch granules, and phytoliths.

Botanical macroremains include archaeological plant remains visible to the naked eye, though usually analyzed with microscopes. These include various types of seeds, fruits, wood, and other large parts of plants.

Botanical microremains include archaeological plant remains that are too small to see with the naked eye and require microscopes to see. These include phytoliths, pollen, and starch granules.

Botanical Macroremains

Macroremains can be preserved in wet, dry, or frozen environments, but under most circumstances, both wood and seeds are preserved when **carbonized**, or incompletely burned to become inorganic. Carbonization may affect the size and shape of seeds and (to a lesser extent) wood, but often results in the conversion of organic materials into inorganic materials that are still matchable by shape to the type of plant from which they originated. For example, wheat and rice, which produce distinctively shaped grains, still look different from one another after carbonization and can be identified accurately. Identification of samples containing carbonized remains is one of the primary methods by which paleoethnobotanists investigate plant use in the past.



Images of modern, uncarbonized wheat (top-left) and rice (top-right) grains; archaeological carbonized wheat (bottom-left) and rice (bottom-right) are still identifiable.

Source: Dr. Marston

Carbonization is the process by which heat converts organic plant materials into inorganic carbon, such as converting wood into charcoal. Carbonization occurs under low heat over long periods or at high heat for short periods, typically in the presence of limited oxygen. With ample oxygen, plant materials are more likely to burn entirely into ash than to carbonize.

Botanical Microremains

Microremains include **phytoliths**, **pollen**, and **starch granules**. These are preserved under different conditions due to their distinct compositions and both chemical and physical structures.

Phytoliths are inorganic structures of silica (SiO_2) that form within and between the cells of many plants. Phytoliths are durable in a wide variety of archaeological contexts but dissolve under highly alkaline (high-pH) conditions.

Not all plants produce phytoliths; some produce phytoliths throughout the plant, and others, only in specific parts of the plant. Because these structures are formed within and among cells, they adopt characteristic shapes based on the shapes of those cells and thus can be matched to the type of plant from which they originated. They are the most durable type of botanical remains because they are inorganic, often numerous, and produced without carbonization. Phytoliths can thus be recovered from soils nearly anywhere—which can tell us about the plants living in that area—or from archaeological contexts, which can tell us which plants were used, processed, or disposed of in that area of a site.

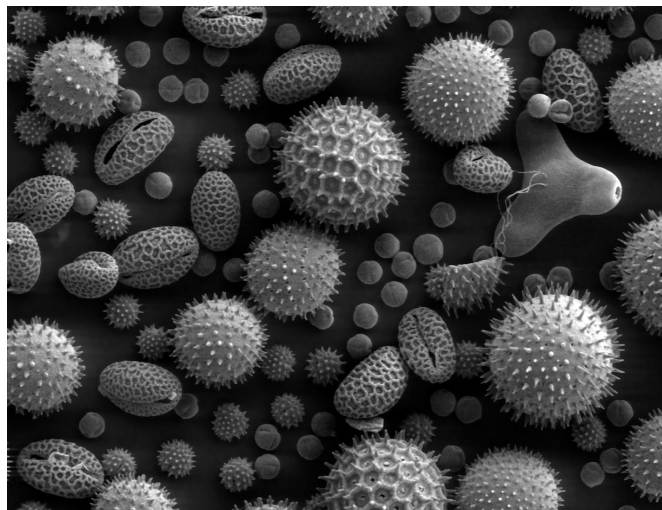


Phytoliths of palm trees (left), maize (center), and squash (right).

Source: Simms, S.R. (2014). Prehispanic Maya foodways: Archaeological and microbotanical evidence from Escalera al Cielo, Yucatan, Mexico [doctoral dissertation, Boston University].
OpenBU: <https://hdl.handle.net/2144/14269>

Pollen is the male sex cell of flowering and seed-bearing plants. The durable outer layer of pollen is degraded by oxygen but is preserved in oxygen-free, waterlogged locations, such as at the bottom of lakes.

Pollen is a durable cell due to its outer layer, termed the exine, which is made from the organic polymer sporopollenin, one of the most inert and chemically resistant molecules known. In fact, it is so resistant to degradation that scientists have long struggled to identify exactly from which smaller molecules it is formed! Pollen exines are preserved under a wide variety of conditions if oxygen is excluded, as oxygen is one of the few ways to degrade sporopollenin. Pollen is thus typically recovered from permanently wet environments (such as the bottoms of lakes) but can also be recovered from soils in archaeological sites. Because pollen is a sex cell, it is produced in large numbers by plants, but some plants produce more than others. Additionally, some plants' pollen is carried by the wind and can travel over long distances, while other plants are pollinated by insects and so do not spread their pollen far from the plant that produced it. Pollen analysts need to take these factors into account when using pollen to infer which plants were present, and in what proportions, in the past.



Pollen comes in an amazing variety of shapes and textures when seen, as here, under scanning electron microscopy. Magnification around 500x.

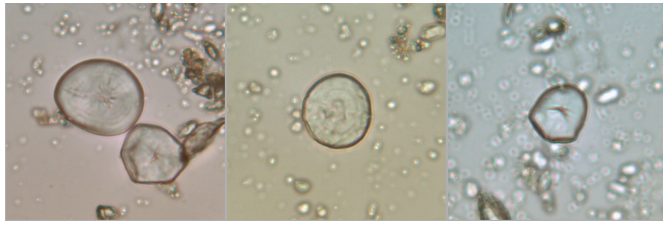
Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Misc_pollen.jpg)

Starch granules are specialized structures used by plants to store starch as a supply of future energy. Starch is often deposited in the stem or root structure of a plant (e.g., a potato) to power future growth, or in a developing seed, to support the new plant when it begins to grow (e.g., a grain of rice). Starch granules are **birefringent** due to their semicrystalline structure of alternating layers of two types of starch: amylose and amylopectin.

Birefringence is an optical property of starch granules (as well as other microscopic structures) due to their semicrystalline structure. Under cross-polarized light, birefringent structures produce a unique cross-shaped pattern, often termed the extinction cross. This feature helps us identify one type of plant's starch granules from those of another plant.

Starch granules are preserved very poorly when left unprotected, as these are a food source for microorganisms. When starch granules are pushed into small cracks in rough stone or ceramic tools, however, they can be preserved for thousands of years. Therefore, most archaeologists recover starch granules not from soils but from artifacts. Starch does not require any special conditions to be preserved on artifacts, although consistently dry and/or cold conditions can be advantageous.



Starch granules of bean (left, at left), maize (left, at right), chile (center), and manioc/tapioca (right). All images taken with normal (nonpolarized) light.

Source: Simms, S.R. (2014). Prehispanic Maya foodways: